Lec 41

① X is a zero-mean Gaussian vector if $\exists$ a matrix A st

$$X = AW \quad \text{where} \quad W = \begin{pmatrix} W_1 \\ \vdots \\ W_m \end{pmatrix} \text{ st } W_i\text{'s are iid}$$

Standard normal RVs.

② If Z is zero-mean Gaussian vector, then AZ is also zero-mean Gaussian vector for any matrix A.

$\Longrightarrow$ If $A = [a_1 \; a_2 \ldots a_n]$ and $Z = \begin{bmatrix} Z_1 \\ \vdots \\ Z_n \end{bmatrix}$

then $AZ = \sum_{i=1}^{n} a_i Z_i$ is a Gaussian random variable.

① If X, Y are jointly Gaussian then $X+Y$ is Gaussian R.V.

$\begin{pmatrix} X \\ Y \end{pmatrix}$ is Gaussian random vector ✓

② If X, Y are Gaussian random variables then

$X+Y$ need not always be Gaussian

③ If X, Y are uncorrelated Gaussian r.vs, it doesn't

imply that X, Y are independent

④ If X, Y are uncorrelated jointly Gaussian r.vs then

X, Y are independent

sian R.v.

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Gaussian

doesn't

nt

r.vs then

$(X, Y) \sim N(0, \sigma_x^2, 0, \sigma_y^2, \rho)$

$$f_{X,Y}(x,y) = \frac{1}{2\pi\sigma_x\sigma_y\sqrt{1-\rho^2}} e^{-\frac{1}{2}\left[\left(\frac{x}{\sigma_x}\right)^2 + \left(\frac{y}{\sigma_y}\right)^2 - \frac{2\rho xy}{\sigma_x\sigma_y}\right]}$$

If X, Y are uncorrelated

$$\rho(X,Y) = 0 = \rho.$$

$$\rho(X,Y) = \frac{Cov(X,Y)}{\sigma_x\sigma_y}$$

$$\Rightarrow f_{X,Y}(x,y) = \frac{1}{2\pi\sigma_x\sigma_y} e^{-\frac{1}{2}\left(\frac{x^2}{\sigma_x^2} + \frac{y^2}{\sigma_y^2}\right)}$$

$$= f_X(x)\, f_Y(y)$$

$$f_X(x) = \frac{e^{-\frac{1}{2}\frac{x^2}{\sigma_x^2}}}{\sqrt{2\pi}\sigma_x} \qquad f_Y(y) = \frac{e^{-\frac{1}{2}\frac{y^2}{\sigma_y^2}}}{\sqrt{2\pi}\sigma_y}$$

$Y = x$

X

$Y|$

$Y|$

$f_Y(y$

Y

$Y = XZ$ where $Z = \{+1, -1\}$

$$Z = \begin{cases} 1 & \text{w.p. } 1/2 \\ -1 & 1/2 \end{cases} \Rightarrow E[Z] = 0$$

$X \sim N(0,1)$. $Z \perp X$

$Y \mid Z = +1 \sim N(0,1)$

$Y \mid Z = -1 \sim N(0,1)$

$$f_Y(y) = \frac{1}{2}\frac{e^{-y^2/2}}{\sqrt{2\pi}} + \frac{1}{2}\frac{e^{-y^2/2}}{\sqrt{2\pi}} = \frac{e^{-y^2/2}}{\sqrt{2\pi}}$$

$Y \sim N(0,1)$

$X \sim N(0,1)$

$Y \sim N(0,1)$

$$X + Y = 0 \, 1_{\{Z=-1\}} + 2X \, 1_{\{Z=1\}}$$

$P(X+Y=0) = \frac{1}{2}$

$\rightarrow \boxed{X+Y}$ can't be Gaussian

Why? Gaussian RV is continuous RV

Can't have prob $1/2$ at 0

$P(X=-y) = 0$

$P(X=-y \mid Y=y) = 1/2$

$\Downarrow$

X, Y are not independent

$$\mathrm{Cov}(X,Y) = E[XY] - E[X]E[Y]$$
$$= E[X X Z] = E[X^2 Z] = E[X^2]E[Z] = 0$$

$$P(X = -y \mid Y = y) = P\left(\frac{Y}{Z} = -y \,\middle|\, Y = y\right) = P(Z = -1 \mid Y = y).$$
$$= P(Z = -1) = 1/2.$$

Lec 41

Suppose X is a zero-mean Gaussian vector s.t.

$X = AW$ $\quad$ W is standard normal vector $\quad$ iid

① Suppose A is $n \times n$ invertible matrix.

$$K_X = E[\underbrace{X}_{n\times 1}\underbrace{X^T}_{1\times n}]$$

$$= E[A W W^T A^T]$$

$$= A\,E[W W^T]\,A^T = A I A^T = A A^T$$

as W_i's are independent

0

$$E[WW^T] = \begin{bmatrix} \underbrace{Cov(W_1,W_1)}_{1} & Cov(W_1,W_2) & & \\ & \ddots & & \\ & & \ddots & \\ & & & Cov(W_n,W_n) \end{bmatrix} = \underbrace{I}_{n\times n}.$$

$$\det(K_X) = \det(A)\det(A^T)$$
$$= (\det(A))^2$$
$$> 0.$$

Any covariance matrix K_X is a p.s.d matrix

$$K_X = E[\bar{X}\bar{X}^T]$$

For any $\underline{a} \in \mathbb{R}^n$ if we show that

$\underline{a}^T K_X \underline{a} \geq 0$ then $\Rightarrow K_X$ is a p.s.d matrix

$$\underline{a}^T K_X \underline{a} = \underline{a}^T E[\bar{X}\bar{X}^T]\underline{a} = E[\underline{a}^T \bar{X}\bar{X}^T \underline{a}]$$

Let $Y = \underbrace{\underline{a}^T}_{1\times n} \underbrace{\bar{X}}_{n\times 1}$

$$= E[YY^T] = E[Y^2] \geq 0$$

Y is random variable

$f_X(x) = f_{X}$

is a p.s.d matrix

$$W \Longleftrightarrow X$$

$$X = AW$$

$$\begin{bmatrix} X_1 \\ \vdots \\ X_n \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ \vdots & & & \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix} \begin{bmatrix} W_1 \\ W_2 \\ \vdots \\ W_n \end{bmatrix}$$

$$X_1 = g_1(W_1 \ldots W_n) = \sum_{i=1}^{n} a_{1i} W_i$$

$$X_j = g_j(W_1, \ldots W_n) = \sum_{i=1}^{n} a_{ji} W_i$$

$$f_X(\underline{x}) = f_{X_1 \ldots X_n}(x_1, \ldots x_n) = \frac{f_W(w_1, \ldots w_n)}{\left| J_{w_1 \ldots w_n}(x_1, \ldots x_n) \right|} \quad \text{st} \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = A \begin{pmatrix} w_1 \\ \vdots \\ w_n \end{pmatrix} \Rightarrow$$

$$\underline{w} = A^{-1} \underline{x}$$

where $J\begin{pmatrix} x_1, \ldots x_n \\ w_1 \ldots w_n \end{pmatrix} = \begin{vmatrix} \frac{\partial g_1}{\partial w_1} & \frac{\partial g_1}{\partial w_2} & \frac{\partial g_1}{\partial w_3} & \cdots & \frac{\partial g_1}{\partial w_n} \\ \vdots & & & & \\ \frac{\partial g_n}{\partial w_1} & - & - & - & \frac{\partial g_n}{\partial w_n} \end{vmatrix}$

$= \begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ \vdots & & & \\ a_{n1} & \cdots & & a_{nn} \end{vmatrix}$

$= |A|$

$|A| = |K_X|^{1/2}$

$\begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = A \begin{pmatrix} w_1 \\ \vdots \\ w_n \end{pmatrix} \Rightarrow f_X(\underline{x}) = \dfrac{f_W(A^{-1}\underline{x})}{|K_X|^{1/2}}$

$\underline{w} = A^{-1}\underline{x}$

$f_W(\underline{w}) =$ ...

$=$ ...

$=$ ...

$f_X(\underline{x}) =$ ...

$$= \begin{vmatrix} \frac{\partial g_1}{\partial w_1} & \frac{\partial g_1}{\partial w_2} & \frac{\partial g_1}{\partial w_3} & \cdots & \frac{\partial g_1}{\partial w_n} \\ \vdots & & & & \\ \frac{\partial g_n}{\partial w_1} & - & - & - & \frac{\partial g_n}{\partial w_n} \end{vmatrix}$$

$$\begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ \vdots & & & \\ a_{n1} & \cdots & & a_{nn} \end{vmatrix}$$

$$|A|$$

$$= \frac{f_W(A^{-1}\underline{x})}{|K_X|^{1/2}}$$

$$f_W(\underline{w}) = \prod_{i=1}^{n} \frac{e^{-\frac{1}{2}w_i^2}}{\sqrt{2\pi}}$$

$$= \frac{1}{(\sqrt{2\pi})^n} e^{-\frac{1}{2}\sum_{i=1}^{n} w_i^2}$$

$$= \frac{1}{(2\pi)^{n/2}} e^{-\frac{1}{2}\underline{w}^T\underline{w}}$$

$$\begin{bmatrix} w_1 & \cdots & w_n \end{bmatrix} \begin{bmatrix} w_1 \\ \vdots \\ w_n \end{bmatrix}$$

$$K_X^{-1} \; ?$$

$$f_X(\underline{x}) = \frac{f_W(A^{-1}\underline{x})}{|K_X|^{1/2}} = \frac{1}{\left((2\pi)^n|K_X|\right)^{1/2}} e^{-\frac{1}{2}\underline{x}^T (A^{-1})^T A^{-1}\underline{x}}$$

$$K_X = AA^T \quad K_X^{-1} = (A^T)^{-1}A^{-1}$$

Lec 41

Given covariance matrix K_X of Gaussian vector X,

if K_X is invertible then

$$f_X(\underline{x}) = \frac{e^{-\frac{1}{2} \underline{x}^T K_X^{-1} \underline{x}}}{(\sqrt{2\pi})^n |K_X|^{1/2}}$$

Given K_X is invertible $\Rightarrow$ eigen values are > 0

K_X is psd $\Rightarrow$ $\exists R$ st $K_X = RR^T$

Claim: R is invertible, if not then K_X is not invertible

Examples:

(i) $K_X = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ $\quad A = I \Rightarrow K_X = AA^T$

(ii) $K_X = \begin{bmatrix} 4 & 0 \\ 0 & 2 \end{bmatrix}$ $\quad A = \begin{bmatrix} 2 & 0 \\ 0 & \sqrt{2} \end{bmatrix}$

(iii) $K_X = \begin{bmatrix} 1 & -0.5 \\ -0.5 & 1 \end{bmatrix}$.

(iv) $K_X = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$

(iii) $K_X = QDQ^T$ then define $A = QD^{1/2}Q^T$

$(1-\lambda)(1-\lambda) - \frac{1}{4} = 0 \Rightarrow 1 - 2\lambda + \lambda^2 + 3/4 = 0$

$(\lambda - 3/2)(\lambda - \frac{1}{2}) = 0$

A^T

$$\begin{bmatrix} 1 & -0.5 \\ -0.5 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \frac{3}{2} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$x_1 - \frac{x_2}{2} = \frac{3}{2} x_1 \Rightarrow x_1 = -x_2$$

$\frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$ is an eigen vector corresponding to eigen value $3/2$

$$\begin{bmatrix} 1 & -0.5 \\ -0.5 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$x_1 - 0.5 x_2 = \frac{1}{2} x_1$$

$$\Rightarrow x_1 = x_2$$

$\frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ is eigen vector corresponding to eigen value $1/2$

$/4 = 0$

$) \left(\lambda - \frac{1}{2}\right) = 0$

$= -x_2$

...ctor corresponding to
...igen value $3/2$

$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$

$\frac{1}{2} x_1$

...vector corresponding to
eigen value $1/2$

$$Q = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$$

$$D = \begin{bmatrix} 3/2 & \\ & 1/2 \end{bmatrix}$$

$$\Downarrow$$

$$K_x = Q D Q^T$$

$$A = Q D^{1/2} Q^T$$

$$= \frac{1}{2} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} \sqrt{3/2} & \\ & \sqrt{\frac{1}{2}} \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$$

$$= \frac{1}{2} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} \sqrt{3/2} & -\sqrt{3/2} \\ \sqrt{1/2} & \sqrt{1/2} \end{bmatrix}$$

$$= \frac{1}{2} \begin{bmatrix} \frac{\sqrt{3}+1}{\sqrt{2}} & \frac{-\sqrt{3}+1}{\sqrt{2}} \\ \frac{-\sqrt{3}+1}{\sqrt{2}} & \end{bmatrix}$$